

Elif Nebioglu

Paris, France

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Education

ENS Paris-Saclay

Paris, France

Master MVA (Mathematics, Vision, Apprentissage), GPA: 3.7 / 4.0 (Honor Student)

2024–2025

Selected Coursework: Convex Optimization, Object Recognition and Computer Vision, Reinforcement Learning, Stopping Times and Online Algorithms, Geometric Data Analysis, Probabilistic Graphical Models, Representation Learning for Medical Imaging, Bayesian ML, Deep Learning in Practice, Generative Modelling, Graphs in ML.

TOBB University of Economics and Technology

Ankara, Türkiye

Bachelors of Science in Computer Engineering, GPA: 3.21 / 4.0 (Honor Student)

2019–2024

Selected Coursework: Object Oriented Programming, Data Structures and Algorithms, Design and Analysis of Algorithms, Combinatorics and Graph Theory, Computer Vision, Deep Learning, Introduction to Machine Learning, Image Processing.

Publications

2026: Nebioglu, E.*, Bilgiç, E.*, Popescu, A. *AI-Generated Image Detectors Overrely on Global Artifacts: Evidence from Inpainting Exchange*. arXiv:2602.00192, 2026 (under review). *Equal contribution.

Introduced INP-X, a diagnostic method exposing shortcut learning in AI-generated image detectors by removing global VAE artifacts. Evaluated 13 detectors on a 90K-image benchmark, showing accuracy drops from 91% to ~55%.

Work Experience

Corma

Paris, France

AI Engineer

March 2026–Present

- Built a production multi-agent system using LangGraph and remote browser automation to handle SaaS provisioning, deprovisioning, and license extraction across 9+ platforms (Slack, Notion, Miro, Cursor, Figma, Google, etc.).
- Designed an end-to-end automation builder pipeline with a chat supervisor, per-block eval harness, self-healing post-run watcher, and SQLite-backed state management.
- Developed an LLM-based eval framework for autonomous agent behavior, covering prompt injection, cross-company data leakage, multi-capability reasoning, and out-of-scope detection.
- Investigated detectability of LLM-based autonomous agents, contributing to R&D initiatives including a France 2030 grant proposal.

Supervised by Adrian Popescu (Université Paris-Saclay)

Paris, France

Independent Researcher

Nov 2025–February 2026

- Conceived an independent research project on shortcut learning in AI-generated image detectors, from initial hypothesis through experimental design, dataset construction, and manuscript preparation.
- Resulting paper currently under review at ICML.

DxO Labs

Paris, France

Computer Vision Research Intern

Apr 2025–Oct 2025

- Designed and implemented an end-to-end face analysis pipeline for portrait retouching.
- Integrated state-of-the-art face detection and segmentation models.
- Developed custom facial warping algorithms in Python.
- Built high-quality facial masks using matting and post-processing techniques.
- Collaborated with R&D engineers to integrate solutions into DxO PhotoLab.

TOBB University of Economics and Technology

Ankara, Türkiye

Computer Vision Research Intern

Jul 2024–Sep 2024

- Developed an end-to-end video-to-label machine learning pipeline to analyze and classify stereotypic locomotor behaviors in fruit flies (*Drosophila*) within a TÜBİTAK-funded cognitive neuroscience project.
- Implemented video preprocessing, feature extraction, and deep learning-based classification models, and optimized the system for reliable behavioral analysis.

TEMSA

Istanbul, Türkiye

Data Science Intern

Jul 2024–Sep 2024

- Designed and implemented an anomaly-detection system for the MD-9 electric bus.
- Cleaned and integrated vehicle CAN telemetry data, analyzed multivariate signal patterns.
- Trained time-series models to learn normal operating behavior and automatically flag anomalies for predictive maintenance.

STM

Ankara, Türkiye

Software Engineering Intern

Oct 2023–Jan 2024

- Worked as a full-stack developer, contributing to in-house web applications using Angular and C#.

Telecom Paris

Paris, France

Research Intern

Feb 2023–May 2023

- Contributed to ACMoM's Eclipse-based model transformation benchmark.
- Automated experiment execution, optimized workflow scripts, and collected and reported performance metrics for systematic evaluation.

Projects

Text-to-3D Human Motion Generation (Reproduction & Extensions) – MVA: Object Recognition & Computer Vision: Reproduced the Motion Diffusion Model (MDM) with SMPL-based motion outputs on HumanML3D and KIT-ML datasets. Extended the framework with TMR++ text augmentation, cross-dataset training, and FPS variation, yielding improved generalization and retrieval performance across datasets.

Plug-and-Play Split Gibbs Sampler for Image Restoration – MVA: Generative Modelling: Implemented and extended a Plug-and-Play Split Gibbs Sampling (PnP-SGS) framework for Bayesian image deblurring and restoration. Compared posterior sampling using learned denoisers with direct DDPM-based generation, and investigated alternative prior models to analyze reconstruction quality and uncertainty.

VibeTune – Bachelor's Thesis: Designed and developed a prompt-driven music recommendation system by fine-tuning the Mistral-7B large language model and integrating ChatGPT-based sentiment analysis. Built an interactive React-based web interface to enable personalized and context-aware music discovery.

Teaching Experience

TOBB University of Economics and Technology: In the Linear Algebra Applications Laboratories, assisted 80 students.

Private Tutoring: Taught Physics, Mathematics to middle and high school students.

Technical Skills

Languages: Python, Java, C++, MATLAB, SQL, Verilog

ML/CV: PyTorch, Keras, Scikit-learn, OpenCV, NumPy, Pandas

Tools: Git, Linux

Honors and Awards

The Business World Scholarship: Merit-based scholarship covering five years of tuition and living expenses at TOBB University, awarded by the Istanbul Chamber of Industry for ranking in the top 0.003 in the National University Examination.

Other

Languages: Turkish (C2), English (C1), French (B1), German (B1)

Interests: Swimming (10 years), Theatre, Psychology, Tennis, Oil Painting.

References

Adrian Popescu: Professor, Université Paris-Saclay, adrian.popescu@cea.fr

Thomas Veit: Research Scientist, DxO Labs, tveit@dxo.com

Ahmet Murat Özbayoğlu: Professor, TOBB University of Economics and Technology, mozbayoglu@etu.edu.tr

Jean-Emmanuel Deschaud: Lecturer, ENS Paris-Saclay / Mines Paris, jean-emmanuel.deschaud@minesparis.psl.eu